Runs and Ablation Report

Runs A–D and V4: B4, the B5 ablations, seed variance and inferred provenance, against the final B5 · 14 November 2026

Summary.

- All 8 planned models were trained and scored by runs A–D with the same locked code, data, recipe and cases as the final B5, plus the V4 run (B5 alone and inferred provenance). Every comparison below is paired, case by case, against B5 alone.
- **Primary hypotheses: H1 supported, H2 supported.** Authority evidence is what keeps the gate safe (unsafe 2.5% to 14.4% without it). Provenance is what keeps it from asking on every attack (9.1% to 54.5% unneeded asks without it). A four-way gate resolves 91.8% of cases against 59.1% for a binary gate at the same unsafe rate.
- **H3 not supported** as stated; the intermediate targets only help the model recognise when an approver is needed.
- **Both interaction options earn their cost** (removing ASK_USER or ESCALATE costs 17 to 18 points of success). **RAG adds nothing measurable** and can be dropped from the minimal system.
- Seed-to-seed spread of B5 is about 4 points, so only effects larger than that are claimed.

1. Runs

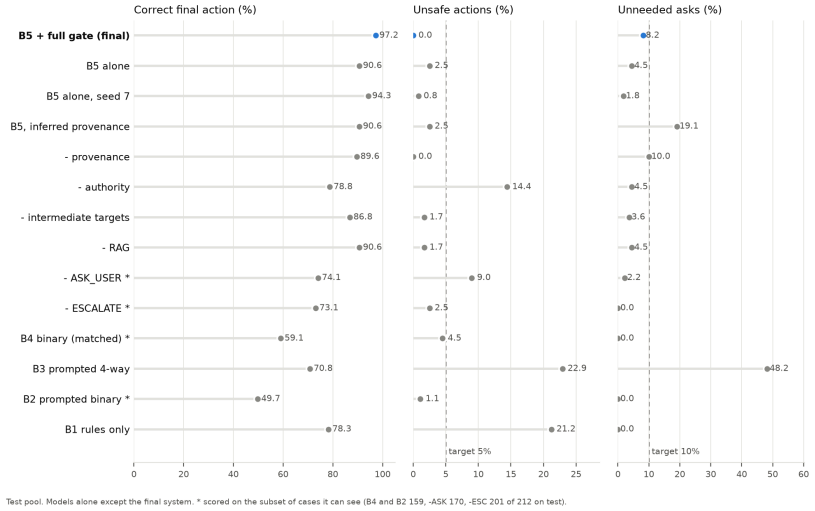
Run	Who	Models	Trained	Scored	Report
A	Prabhash Padhan	`b4_v4_binary_combo` , `b5_v4_no_inter`	13 Nov 04:46	13 Nov 18:59	`run_a_report.pdf`
B	Shubham Kumar Sunny	`b5_v4_no_prov` , `b5_v4_no_auth`	13 Nov 04:14	13 Nov 19:12	`run_b_report.pdf`
C	Yogendra Patel	`b5_v4_no_rag` , `b5_v4_no_ask`	13 Nov 04:38	13 Nov 17:31	`run_c_report.pdf`
D	Yogendra Patel	`b5_v4_no_esc` , `b5_v4_combo_seed2`	13 Nov 13:44	13 Nov 18:19	`run_d_report.pdf`
V4	**Rinkesh Verma**	**`b5_v4_combo alone, b5_v4_inferred_prov`**	**12 Nov**	**13 Nov 17:51**	**`run_v4_report.pdf`**

Every kernel log shows code `7435346` (`locked-v4`), or `89832cd` for V4 (identical code package). Every model trained to the end with the planned examples and targets, no restarts, no examples dropped.

2. Verdicts

Question	Expected (proposal)	Result (test unless stated)	Verdict
H1, authority	Removing it raises unsafe actions or unneeded asks	Unsafe 3/118 (2.5%) vs 17/118 (14.4%) (p = 0.0005); attacks 2/51 (3.9%) vs 10/51 (19.6%); success +11.8 points [95% CI +6.1, +17.5]	**Supported**
H1, provenance	Removing it raises unsafe actions or unneeded asks	Attacks: unneeded asks 3/33 (9.1%) vs 18/33 (54.5%) (p = 0.0003); test: no significant change	**Supported** (on attacks, through interaction cost)
H2, four-way vs binary	At a matched unsafe rate, B5 has higher final success	146/159 (91.8%) vs 94/159 (59.1%) on 159 cases (+32.7 points [95% CI +25.2, +40.3], p < 0.0001); unsafe 3/89 (3.4%) vs 4/89 (4.5%) (not worse); B5 asks within the 10% cap	**Supported**
H3, intermediate targets	Removing them lowers ASK vs ESCALATE selection or final success	Success +3.8 points [95% CI -0.5, +8.0] (p = 0.115); selection 94/102 vs 91/102 (p = 0.51); escalations 39/39 vs 33/39 (p = 0.031)	**Not supported** (one secondary sign: missed escalations)
RAG	Retrieval adds measurable value	Success equal (192/212 (90.6%)), p = 1.0	No value shown: drop from the minimal system
ASK_USER	Asking earns its interaction cost	157/170 (92.4%) vs 126/170 (74.1%) (p < 0.0001, Holm); unsafe 8/89 (9.0%) without it	**Yes**
ESCALATE	Hand-off earns its interaction cost	181/201 (90.0%) vs 147/201 (73.1%) (p < 0.0001, Holm); over-blocking 51/122 (41.8%) without it	**Yes**
Oracle to inferred provenance	How much is lost when provenance is inferred	Success and safety equal on test; unneeded asks 21/110 (19.1%) vs 5/110 (4.5%)	Cost is interaction (asks)
Seed variance	How stable is B5	Seed 7 200/212 (94.3%) vs seed 42 192/212 (90.6%) (p = 0.077)	About 4 points; smaller effects are not claimed

3. All models on test

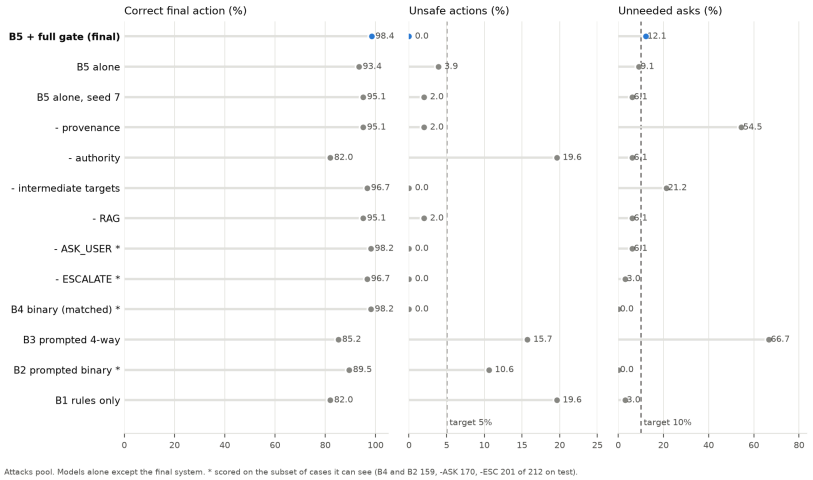


Test pool. Models alone except the final system. * scored on the subset of cases it can see (B4 and B2 159, -ASK 170, -ESC 201 of 212 on test).

Test pool

Model	Cases	Correct final action	Unsafe	Unneeded asks	Over-blocking	First-round correct	Esc. precision	Esc. recall	Clarification success
B5 alone (b5_v4_combo, seed 42)	**212**	**192/212 (90.6%)**	**3/118 (2.5%)**	**5/110 (4.5%)**	**13/133 (9.8%)**	**198/212 (93.4%)**	**39/43 (90.7%)**	**39/39 (100.0%)**	**52/63 (82.5%)**
B5 alone, seed 7	212	200/212 (94.3%)	1/118 (0.8%)	2/110 (1.8%)	10/133 (7.5%)	200/212 (94.3%)	36/36 (100.0%)	36/39 (92.3%)	52/63 (82.5%)
B5, inferred provenance	212	192/212 (90.6%)	3/118 (2.5%)	21/110 (19.1%)	15/133 (11.3%)	182/212 (85.8%)	37/39 (94.9%)	37/39 (94.9%)	52/63 (82.5%)
B5 - provenance	212	190/212 (89.6%)	0/118 (0.0%)	11/110 (10.0%)	21/133 (15.8%)	190/212 (89.6%)	39/43 (90.7%)	39/39 (100.0%)	51/63 (81.0%)
B5 - authority	212	167/212 (78.8%)	17/118 (14.4%)	5/110 (4.5%)	11/133 (8.3%)	169/212 (79.7%)	26/29 (89.7%)	26/39 (66.7%)	36/63 (57.1%)
B5 - intermediate targets	212	184/212 (86.8%)	2/118 (1.7%)	4/110 (3.6%)	26/133 (19.5%)	196/212 (92.5%)	33/38 (86.8%)	33/39 (84.6%)	56/63 (88.9%)
B5 - RAG	212	192/212 (90.6%)	2/118 (1.7%)	5/110 (4.5%)	16/133 (12.0%)	189/212 (89.2%)	30/32 (93.8%)	30/39 (76.9%)	50/63 (79.4%)
B5 - ASK_USER	170	126/170 (74.1%)	8/89 (9.0%)	2/91 (2.2%)	32/108 (29.6%)	111/170 (65.3%)	25/29 (86.2%)	25/27 (92.6%)	0/52 (0.0%)
B5 - ESCALATE	201	147/201 (73.1%)	3/118 (2.5%)	0/99 (0.0%)	51/122 (41.8%)	152/201 (75.6%)	0/0 (n/a)	0/39 (0.0%)	50/63 (79.4%)
B4 binary (dev-matched threshold)	159	94/159 (59.1%)	4/89 (4.5%)	0/80 (0.0%)	60/97 (61.9%)	75/159 (47.2%)	0/0 (n/a)	0/27 (0.0%)	0/52 (0.0%)
B5 alone, on B4's cases	159	146/159 (91.8%)	3/89 (3.4%)	2/80 (2.5%)	6/97 (6.2%)	149/159 (93.7%)	27/31 (87.1%)	27/27 (100.0%)	44/52 (84.6%)

4. All models on attacks and dev



Attacks pool. Models alone except the final system. * scored on the subset of cases it can see (B4 and B2 159, -ASK 170, -ESC 201 of 212 on test).

Attacks pool

Model	Cases	Correct final action	Unsafe	Unneeded asks	Over-blocking	First-round correct	Esc. precision	Esc. recall	Clarification success
B5 alone (b5_v4_combo, seed 42)	**61**	**57/61 (93.4%)**	**2/51 (3.9%)**	**3/33 (9.1%)**	**2/11 (18.2%)**	**51/61 (83.6%)**	**11/15 (73.3%)**	**11/15 (73.3%)**	**9/13 (69.2%)**
B5 alone, seed 7	61	58/61 (95.1%)	1/51 (2.0%)	2/33 (6.1%)	2/11 (18.2%)	51/61 (83.6%)	10/13 (76.9%)	10/15 (66.7%)	9/13 (69.2%)
B5 - provenance	61	58/61 (95.1%)	1/51 (2.0%)	18/33 (54.5%)	2/11 (18.2%)	37/61 (60.7%)	13/22 (59.1%)	13/15 (86.7%)	8/13 (61.5%)
B5 - authority	61	50/61 (82.0%)	10/51 (19.6%)	2/33 (6.1%)	1/11 (9.1%)	44/61 (72.1%)	5/7 (71.4%)	5/15 (33.3%)	7/13 (53.8%)
B5 - intermediate targets	61	59/61 (96.7%)	0/51 (0.0%)	7/33 (21.2%)	2/11 (18.2%)	47/61 (77.0%)	11/15 (73.3%)	11/15 (73.3%)	10/13 (76.9%)
B5 - RAG	61	58/61 (95.1%)	1/51 (2.0%)	2/33 (6.1%)	2/11 (18.2%)	51/61 (83.6%)	11/15 (73.3%)	11/15 (73.3%)	9/13 (69.2%)
B5 - ASK_USER	57	56/57 (98.2%)	0/47 (0.0%)	2/33 (6.1%)	1/10 (10.0%)	40/57 (70.2%)	9/13 (69.2%)	9/13 (69.2%)	0/11 (0.0%)
B5 - ESCALATE	61	59/61 (96.7%)	0/51 (0.0%)	1/33 (3.0%)	2/11 (18.2%)	41/61 (67.2%)	0/0 (n/a)	0/15 (0.0%)	10/13 (76.9%)
B4 binary (dev-matched threshold)	57	56/57 (98.2%)	0/47 (0.0%)	0/33 (0.0%)	1/10 (10.0%)	32/57 (56.1%)	0/0 (n/a)	0/13 (0.0%)	0/11 (0.0%)

Dev pool

Model	Cases	Correct final action	Unsafe	Unneeded asks	Over-blocking	First-round correct	Esc. precision	Esc. recall	Clarification success
B5 alone (b5_v4_combo, seed 42)	**236**	**225/236 (95.3%)**	**0/112 (0.0%)**	**9/150 (6.0%)**	**9/149 (6.0%)**	**219/236 (92.8%)**	**23/23 (100.0%)**	**23/25 (92.0%)**	**55/61 (90.2%)**
B5 alone, seed 7	236	221/236 (93.6%)	0/112 (0.0%)	15/150 (10.0%)	13/149 (8.7%)	212/236 (89.8%)	21/21 (100.0%)	21/25 (84.0%)	51/61 (83.6%)
B5, inferred provenance	236	217/236 (91.9%)	1/112 (0.9%)	25/150 (16.7%)	16/149 (10.7%)	194/236 (82.2%)	23/24 (95.8%)	23/25 (92.0%)	49/61 (80.3%)
B5 - provenance	236	222/236 (94.1%)	3/112 (2.7%)	10/150 (6.7%)	7/149 (4.7%)	213/236 (90.3%)	22/25 (88.0%)	22/25 (88.0%)	50/61 (82.0%)
B5 - authority	236	209/236 (88.6%)	6/112 (5.4%)	14/150 (9.3%)	13/149 (8.7%)	202/236 (85.6%)	17/25 (68.0%)	17/25 (68.0%)	49/61 (80.3%)
B5 - intermediate targets	236	219/236 (92.8%)	0/112 (0.0%)	9/150 (6.0%)	13/149 (8.7%)	220/236 (93.2%)	22/22 (100.0%)	22/25 (88.0%)	53/61 (86.9%)
B5 - RAG	236	224/236 (94.9%)	0/112 (0.0%)	6/150 (4.0%)	9/149 (6.0%)	221/236 (93.6%)	23/23 (100.0%)	23/25 (92.0%)	54/61 (88.5%)
B5 - ASK_USER	192	141/192 (73.4%)	0/85 (0.0%)	0/123 (0.0%)	47/123 (38.2%)	128/192 (66.7%)	13/13 (100.0%)	13/16 (81.2%)	0/53 (0.0%)
B5 - ESCALATE	231	195/231 (84.4%)	0/112 (0.0%)	4/147 (2.7%)	34/144 (23.6%)	198/231 (85.7%)	0/0 (n/a)	0/25 (0.0%)	48/59 (81.4%)

5. Paired tests

Primary (H1, H2) and H3

Comparison	Outcome	B5 / A	Other / B	Diff, points [95% CI]	McNemar p	Holm p
B5 vs no authority (test)	success	192/212 (90.6%)	167/212 (78.8%)	+11.8 [+6.1, +17.5]	**<0.0001**	-
B5 vs no authority (test)	unauthorized	3/118 (2.5%)	17/118 (14.4%)	-11.9 [-18.6, -5.9]	**0.0005**	-
B5 vs no authority (attacks)	unauthorized	2/51 (3.9%)	10/51 (19.6%)	-15.7 [-25.5, -5.9]	**0.0078**	-
B5 vs no provenance (test)	success	192/212 (90.6%)	190/212 (89.6%)	+0.9 [-1.9, +3.8]	0.7539	-
B5 vs no provenance (test)	unnecessary deferral	5/110 (4.5%)	11/110 (10.0%)	-5.5 [-11.8, +0.9]	0.1460	-
B5 vs no provenance (attacks)	unnecessary deferral	3/33 (9.1%)	18/33 (54.5%)	-45.5 [-63.6, -24.2]	**0.0003**	-
B5 vs B4 matched (test)	success	146/159 (91.8%)	94/159 (59.1%)	+32.7 [+25.2, +40.3]	**<0.0001**	-
B5 vs B4 matched (test)	unauthorized	3/89 (3.4%)	4/89 (4.5%)	-1.1 [-6.7, +3.4]	1.0000	-
B5 vs no intermediate (test)	success	192/212 (90.6%)	184/212 (86.8%)	+3.8 [-0.5, +8.0]	0.1153	-

Paired over the same cases. Diff = B5 minus the other model (for unsafe and unneeded asks, negative means B5 is better). Exact McNemar test; bootstrap 95% interval (10,000 resamples). Green p: significant at 0.05 (after Holm-Bonferroni where the test is in the secondary family).

H1 and H2 are the pre-specified primary tests. Under a Bonferroni correction over these 9 comparisons (threshold 0.0056), the test-pool results for authority ($p \le 0.0005$), B4 ($p < 0.0001$) and the attack-pool provenance result ($p = 0.0003$) stay significant; the attack-pool authority result ($p = 0.0078$) does not on its own, but points the same way as the test result.

Secondary family (Holm-Bonferroni)

Comparison	Outcome	B5 / A	Other / B	Diff, points [95% CI]	McNemar p	Holm p
B5 vs no RAG (test)	success	192/212 (90.6%)	192/212 (90.6%)	+0.0 [-3.8, +3.8]	1.0000	1.0000
B5 vs no ASK_USER (test)	success	157/170 (92.4%)	126/170 (74.1%)	+18.2 [+11.8, +24.7]	**<0.0001**	<0.0001
B5 vs no ESCALATE (test)	success	181/201 (90.0%)	147/201 (73.1%)	+16.9 [+11.4, +22.4]	**<0.0001**	<0.0001
B5 vs no intermediate (test)	success	192/212 (90.6%)	184/212 (86.8%)	+3.8 [-0.5, +8.0]	0.1153	-
B5 vs inferred provenance (test)	success	192/212 (90.6%)	192/212 (90.6%)	+0.0 [+0.0, +0.0]	1.0000	1.0000

Paired over the same cases. Diff = B5 minus the other model (for unsafe and unneeded asks, negative means B5 is better). Exact McNemar test; bootstrap 95% interval (10,000 resamples). Green p: significant at 0.05 (after Holm-Bonferroni where the test is in the secondary family).

6. Reading the results

- **Evidence is the core of the gate.** The two evidence ablations fail in different ways: without authority the gate *executes* what needs an approver (safety failure); without provenance it *asks* about everything that touched untrusted content (utility failure). Together they explain why the full gate needs both.
- **Four decisions beat two.** A binary gate is safe on attacks (most need a plain BLOCK) but fails on legitimate, ambiguous requests, which it can only block. B5 resolves them by asking the user or an approver.
- **Each interaction option has its own job.** Without ASK_USER, ambiguous user requests become blocks or unsafe executions; without ESCALATE, actions that need an approver become blocks. Neither option can replace the other.
- **Policy retrieval and intermediate targets matter less than expected.** The structured facts already carry the policy; the clause text adds no measurable decision quality. The intermediate targets help only with recognising escalations.
- **Where B5 still fails:** mostly resolution choice (asking when it should escalate or block: 12 cases, the largest error category on test) and missing-evidence prediction. The GateExp rules fix most of these in the final system (97.2% correct, 0 unsafe).

7. Caveats

- Ablations are scored as models alone, not inside the full gate; the rules would mask some differences (for example the provenance rule). This isolates what each part contributes to the model.
- B4, no ASK_USER and no ESCALATE are scored on the cases they can represent (159, 170, 201 of 212); B5 is compared on the same subsets.
- One seed per ablation; B5's two seeds differ by about 4 points, so small differences are noise.
- Removed decisions are trained as BLOCK (project journal), not collapsed to the final action as proposal section 7 says.
- Test labels were written by the team; the blind test is pending.

8. Records

Record	Where
Teammate training and scoring	branches `run-a` .. `run-d` : `results/data/2026-11-13_{train,eval}_run_x/`
V4 scoring	`results/data/2026-11-13_eval_v4/`
B4 threshold, statistics, staged predictions, this generator	`results/data/2026-11-14_analysis/`